



Southeast University

Introduction to Software Engineering

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Outline

Part 0. Overview

Part 1. Software Processes

Part 2. Modeling

❖ **Chap 7. Understanding Requirements**

❖ **Chap 8-10. Requirements Modeling**

❖ **Chap 11. Design Concepts**

❖ **Chap 12-14. Design Methods**

Part 3. Quality Assurance

Part 4. Software Project Management

Chap 10. Behavioral Modeling

❖ 1. Behavioral Modeling

- State Diagram
- Sequence Diagram

- 基于行为建模（列出系统的状态、表面系统的状态如何转变、画状态图或顺序图）

■ 创建行为模型的步骤

Behavioral Modeling

❖ **The behavioral model indicates how software will respond to external events or stimuli. To create the model, the analyst must perform the following steps:**

- **Evaluate all use-cases to fully understand the sequence of interaction within the system.**
- **Identify events that drive the interaction sequence and understand how these events relate to specific objects.**
- **Create a sequence for each use-case.**
- **Build a state diagram for the system.**
- **Review the behavioral model to verify accuracy and consistency.**

• 行为模型指示软件如何响应外部事件或刺激。为了创建模型，分析员必须执行以下步骤：

- 评估所有用例以充分理解系统内的交互顺序。
- 确定驱动交互顺序的事件，并了解这些事件如何与特定对象相关。
- 为每个用例创建一个顺序。
- 为系统构建状态图。
- 审查行为模型以验证其准确性和一致性。

Behavioral Modeling

- ❖ **make a list of the different states of a system (How does the system behave?)**
- ❖ **indicate how the system makes a transition from one state to another (How does the system change state?)**
 - **indicate event**
 - **indicate action**
- ❖ **draw a state diagram or a sequence diagram**

- 列出系统的不同状态（系统如何表现？）

- 指示系统如何从一种状态过渡到另一种状态（系统如何改变状态？）

- 指示事件

- 指示动作

- 绘制状态图或顺序图

■ 状态表现

State Representations

- ❖ In the context of behavioral modeling, two different characterizations of states must be considered:
 - the state of each class as the system performs its function and
 - the state of the system as observed from the outside as the system performs its function.
- ❖ The state of a class takes on both passive and active characteristics [CHA93].
 - A *passive state* is simply the current status of all of an object's attributes.
 - The *active state* of an object indicates the current status of the object as it undergoes a continuing transformation or processing.

■ 在行为建模的上下文中，必须考虑两种不同的状态表征： ■ 类状态具有被动和主动两种特征

1. 系统执行其功能时每个类的状态
2. 系统执行其功能时从外部观察到的系统状态

- 被动——对象所有属性的当前状态
- 主动——对象在进行持续的转换或处理时的当前状态

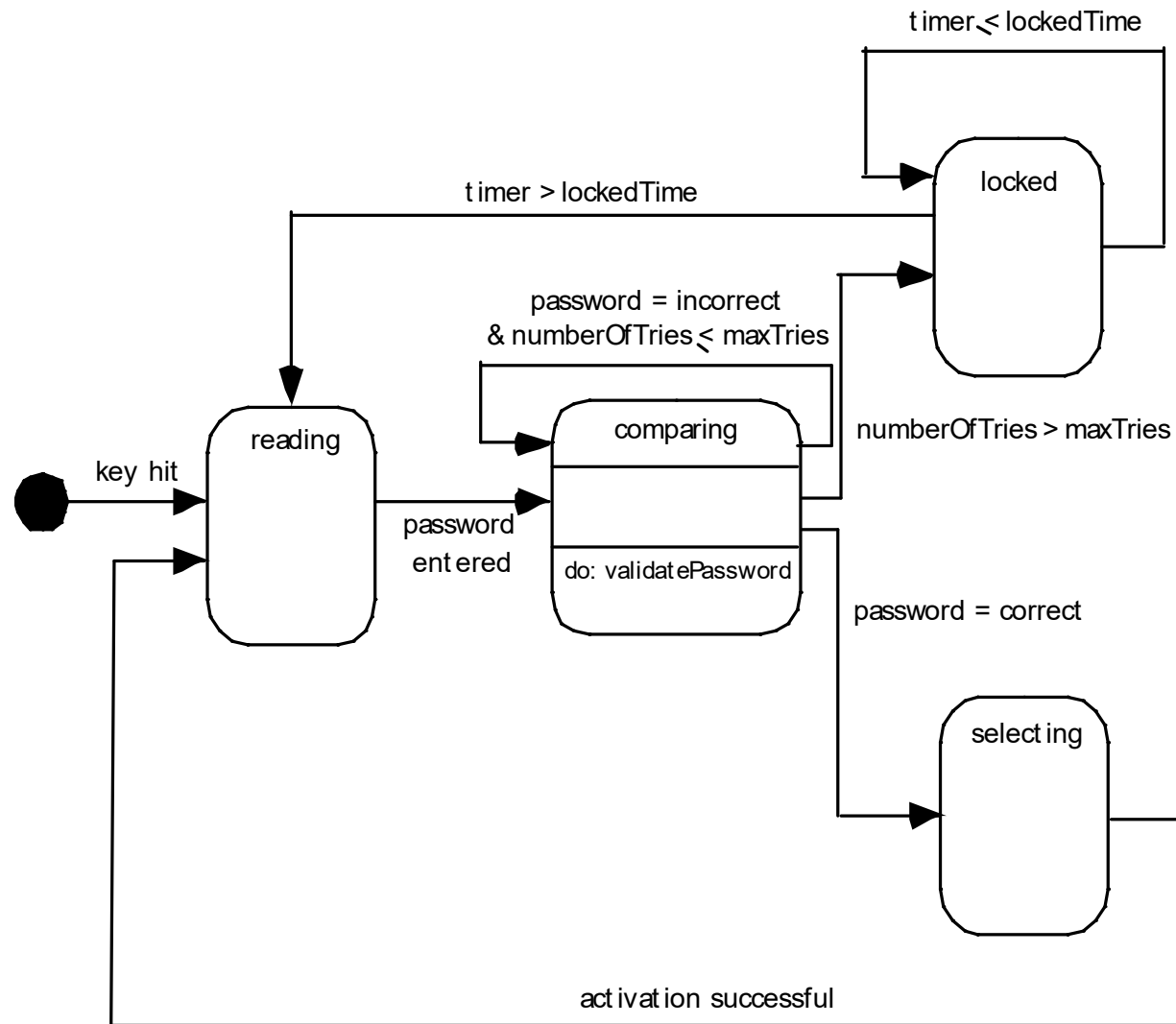
■ 系统的状态

The States of a System

- ❖ **state**—a set of observable circumstances that characterizes the behavior of a system at a given time
- ❖ **state transition**—the movement from one state to another
- ❖ **event**—an occurrence that causes the system to exhibit some predictable form of behavior
- ❖ **action**—process that occurs as a consequence of making a transition

- state——一组可观察到的情况，表征系统在给定时间的行为
- state transition——从一个状态到另一个状态的移动
- event——导致系统表现出某种可预测的行为形式的事件
- action——发生的过程作为过渡的结果

State Diagram for the ControlPanel Class



Sequence Diagram

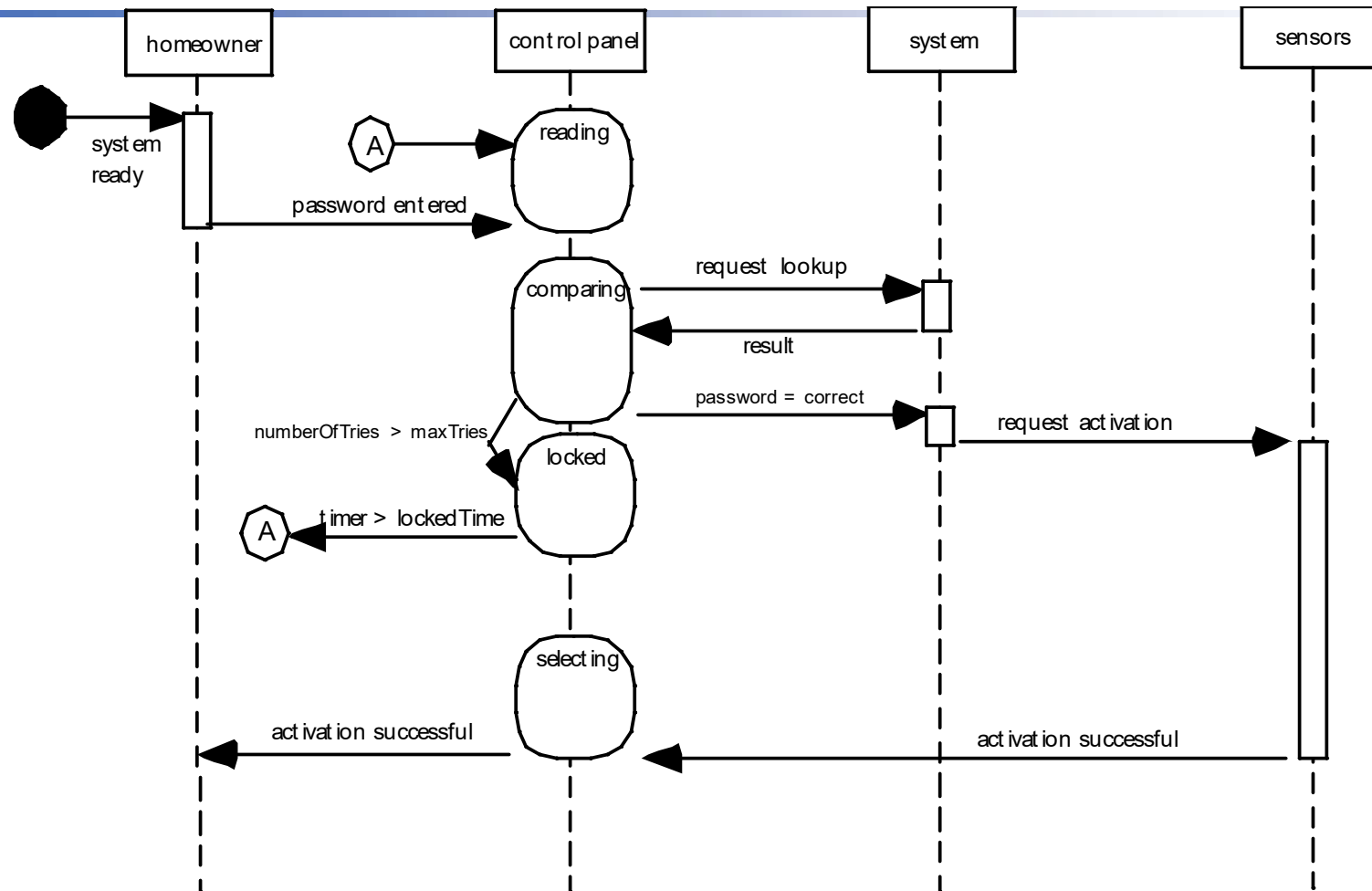
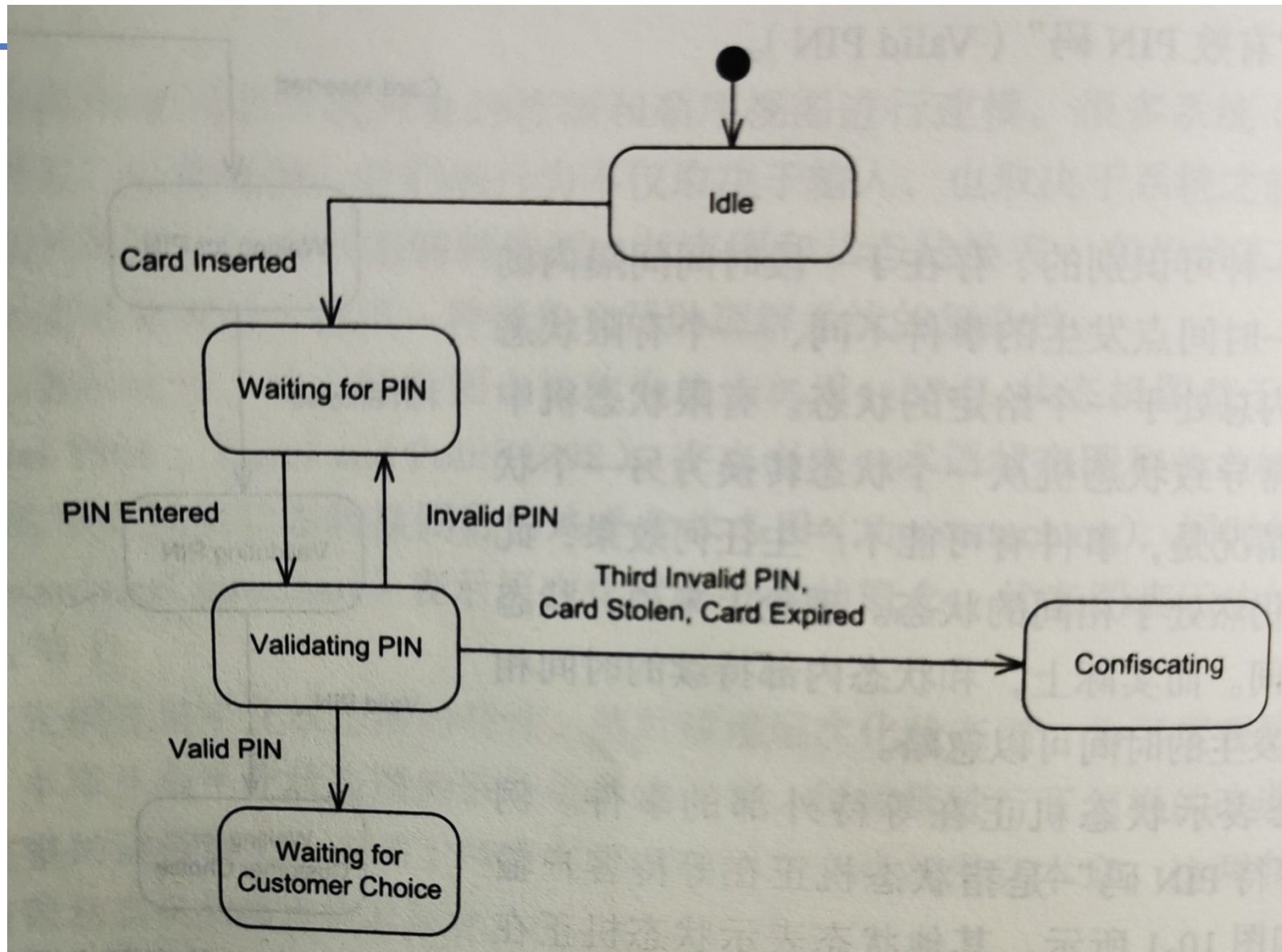


Figure 8.27 Sequence diagram (partial) for *SafeHome* security function

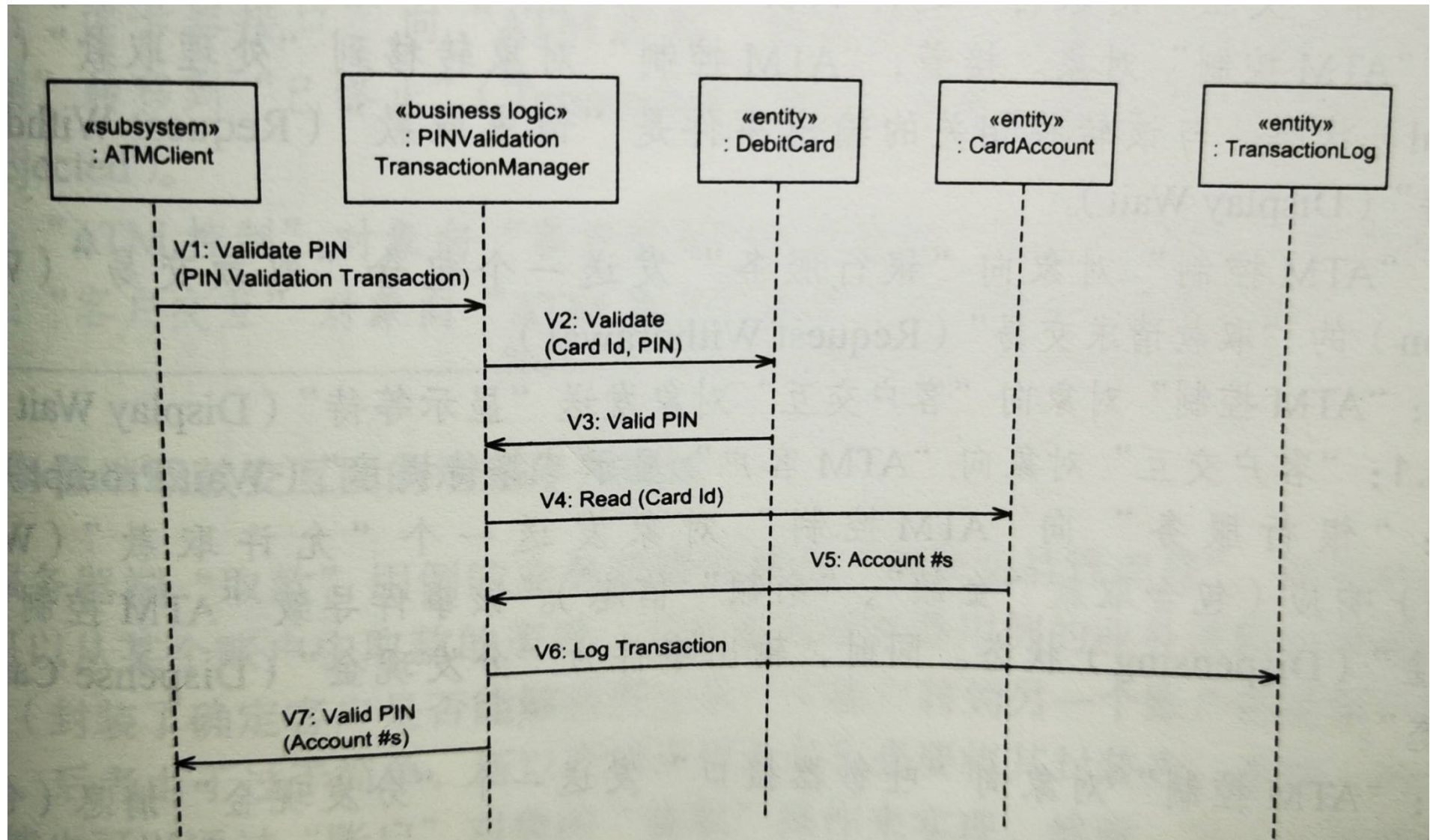
状态图示例

- ❖ 为一个ATM机的工作构建状态图。
 - ATM初始是**空闲状态 (Idle)**，客户将卡片插入ATM机，ATM检测到“**卡片已插入**”事件的发生，则**等待PIN码输入**，用户输入**Pin**码后，**ATM机验证PIN码**，同时**验证该卡是否已挂失或被盗**。若**PIN码验证成功**，则**ATM等待用户选择操作（存/取款）**。若**3次PIN码验证错误**，**没收该卡片**。



顺序图示例

- ❖ 上一个场景中，假设有如下对象：
 - ATM客户
 - 系统的客户交互对象（主业务逻辑）
 - 银行卡
 - 银行账户
 - 交易日志
- ❖ 试画出上述场景的顺序图



Writing the Software Specification



Specification

- ❖ **Correctness**
- ❖ **Integrity**
- ❖ **No ambiguity**
- ❖ **Consistency**
- ❖ **Verifiable**
- ❖ **Understandability**
- ❖ **Modifiable**
- ❖ **Traceable**

3. 软件规约的特性

- 正确性
- 完整性
- 无二义
- 一致性
- 可验证
- 非计算机专家可理解
- 可修改
- 可跟踪



Southeast University



Thank You !

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